

Single VO₂ Neuristor Simulations

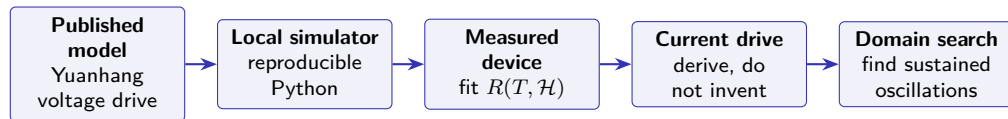
From Yuanhang mechanism validation to specimen parameter estimation

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The Central Research Question



Research question

Can the original electrothermal and hysteretic mechanism produce robust relaxation oscillations under an *ideal imposed current*, and where is that operating domain?

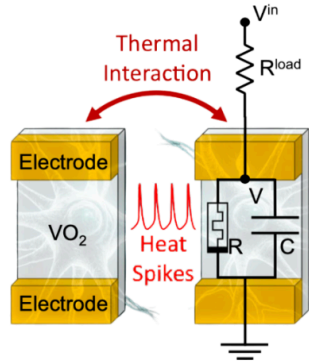
- Preserve the original VO₂ resistance mechanism.
- Reject startup overshoot and timestep-dependent artifacts.
- Visualize internal temperature and resistance dynamics.

Starting Point: Electrical Dynamics

Yuanhang voltage-source node equation

$$C \frac{dV}{dt} = \frac{V_{in} - V}{R_{series}} - \frac{V}{R_{VO_2}}$$

- $C dV/dt$: charge stored at the neuristor node.
- $(V_{in} - V)/R_{series}$: current delivered by the source.
- V/R_{VO_2} : nonlinear leakage through VO_2 .
- $R_{VO_2} = R(T, \mathcal{H})$: temperature and memory couple the electrical and thermal problems.



Voltage-source equivalent circuit.

Starting point: Y.-H. Zhang et al., Nature Communications (2024).

The Feedback Mechanism: Thermal Dynamics

Single-neuristor reduction

$$C_{th} \frac{dT}{dt} = \underbrace{\frac{V^2}{R_{VO_2}}}_{\text{Joule heating}} - \underbrace{S_e(T - T_0)}_{\text{cooling}} + \underbrace{\sigma \eta(t)}_{\text{optional noise}}$$

- C_{th} : thermal inertia of the active device.
- S_e : thermal conductance to substrate/environment.
- Heating lowers R_{VO_2} ; lower resistance changes the voltage and power.
- This delayed negative feedback enables relaxation oscillations.

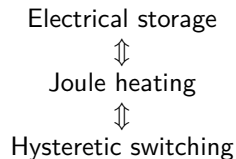
For one isolated device, the inter-neuristor thermal coupling term is zero.

Two coupled state variables

$$V(t), \quad T(t)$$

Material memory

$$\mathcal{H} = \{\delta, T_r, g_r, T_{pr}, \dots\}$$



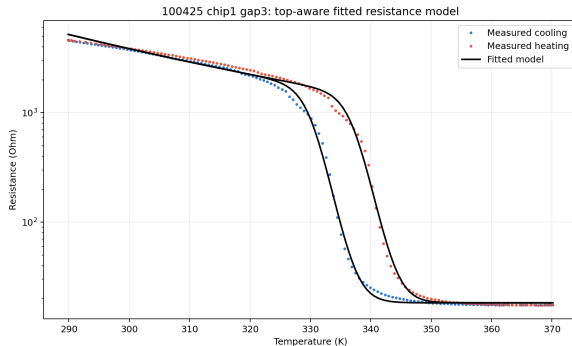
Measured Hysteretic Resistance

$$R(T, \mathcal{H}) = R_0 e^{(E_a/k_B)/T} \underbrace{g(T, \mathcal{H})}_{\text{insulating fraction}} + R_m$$

$$g = \frac{1}{2} + \frac{1}{2} \tanh \left[\beta \left(\delta \frac{w}{2} + T_c - (T + T_p) \right) \right]$$

- T_c, w, β : center, width, and steepness.
- δ : heating or cooling direction.
- T_p : minor-loop shift after a reversal.
- The same law is used in both source configurations.

Hysteresis: L. A. L. de Almeida et al., Optical Engineering (2002).



Sample 100425 chip1 gap3; log-space $R^2 \approx 0.9987$.

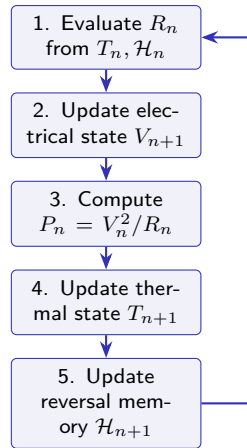
What Happens at Every Timestep?

$$R_n = R(T_n, \mathcal{H}_n)$$

$$V_{n+1} = V_n + \frac{\Delta t}{C} \left(I_{\text{src},n} - \frac{V_n}{R_n} \right)$$

$$T_{n+1} = T_n + \frac{\Delta t}{C_{th}} \left[\frac{V_n^2}{R_n} - S_e(T_n - T_0) \right] + \frac{\sigma}{C_{th}} \sqrt{\Delta t} \zeta_n$$

$$I_{\text{src},n} = \begin{cases} (V_{in,n} - V_n)/R_{series}, & \text{voltage drive,} \\ I_{in,n}, & \text{current drive.} \end{cases}$$



Forward Euler; Euler–Maruyama when $\sigma > 0$.

Production sweeps use $\sigma = 0$.

From Voltage Drive to Ideal Current Drive

Original voltage source

$$C \frac{dV}{dt} = \frac{V_{in} - V}{R_{series}} - \frac{V}{R}$$

I_{src} changes with V

Ideal imposed current

$$I_{in} = C \frac{dV}{dt} + \frac{V}{R}$$

$$C \frac{dV}{dt} = I_{in} - \frac{V}{R}$$

Preserved

Yuanhang $R(T, \mathcal{H})$

single thermal state T

Joule heating and cooling

optional stochastic heating

Assumed in the current-source reduction

one VO₂ device/domain

one thermal node

ideal imposed current

no added phase-lag state

The circuit boundary condition changed; the material mathematics did not.

New Problem: The Oscillatory Domain Was Unknown

Physical controls

I_{in}	imposed electrical drive
C	electrical charging time
C_{th}	thermal inertia
S_e	environmental cooling
T_0	starting/ambient temperature

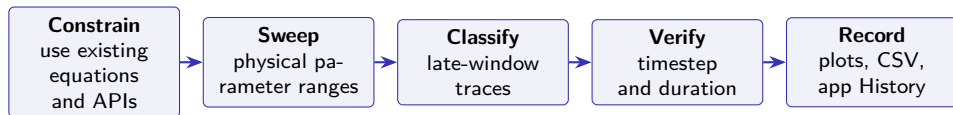
$$\tau_e \sim RC, \quad \tau_{th} \sim \frac{C_{th}}{S_e}$$

Oscillations require enough Joule heating to cross the transition, but enough cooling and electrical recovery to return.

Acceptance criteria

- 1 Oscillations persist in a late-time window.
- 2 More than one full cycle is present.
- 3 Startup overshoot is excluded.
- 4 Frequency is below the numerical/Nyquist limit.
- 5 Period and amplitude survive timestep refinement.
- 6 The path is interpretable on $R(T, \mathcal{H})$.

How I Used Codex as a Research Tool



Representative instruction

“Use the existing current-source simulator without changing the model. Sweep physical values of current, capacitance, thermal capacitance, and thermal conductance. Find sustained oscillations, reject startup overshoot and numerical artifacts, and save successful simulations into app History.”

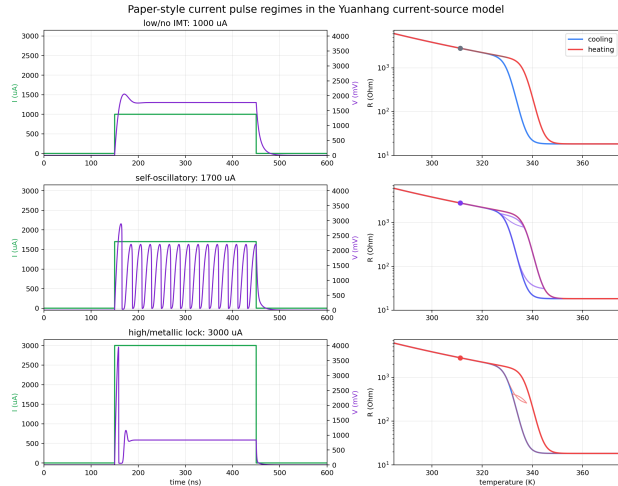
Codex accelerated

- search-script implementation;
- repeated simulations and plots;
- trace metrics and job registration.

Scientific control remained

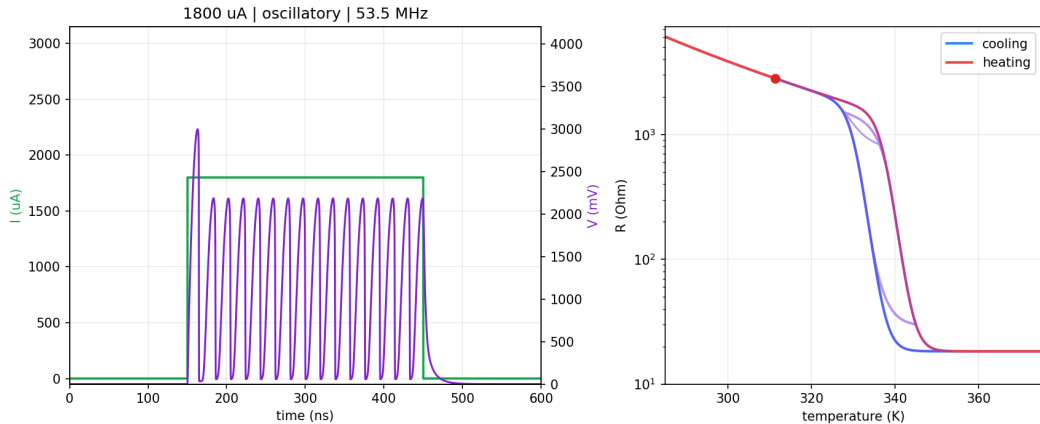
- equations and physical constraints;
- acceptance/rejection criteria;
- interpretation and final claims.

Result: Three Current-Driven Regimes



Low current: insulating equilibrium $\rightarrow$ intermediate current: relaxation oscillation $\rightarrow$ high current: metallic lock.

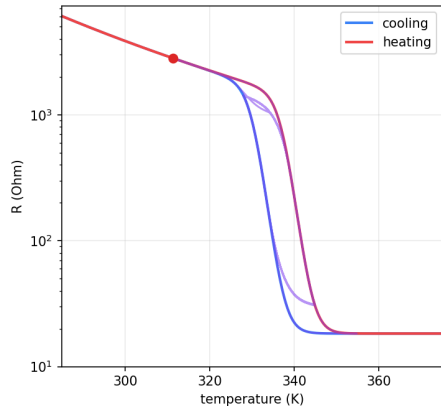
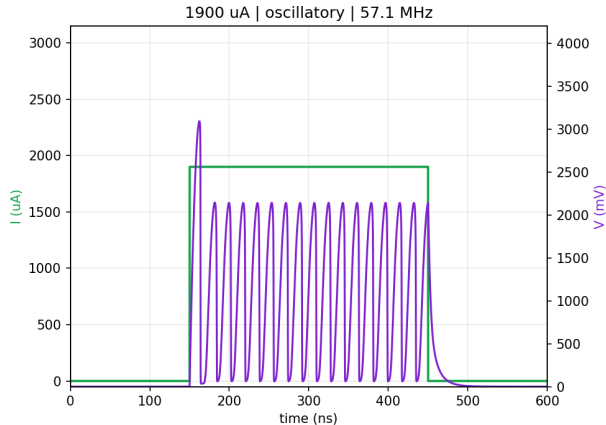
What the Simulation Reveals Inside One Oscillation



1800 μA : sustained 53.5 MHz voltage oscillation. The right panel exposes the otherwise hidden trajectory through the hysteresis loop.

[► Open 10-second time-evolution animation](#)

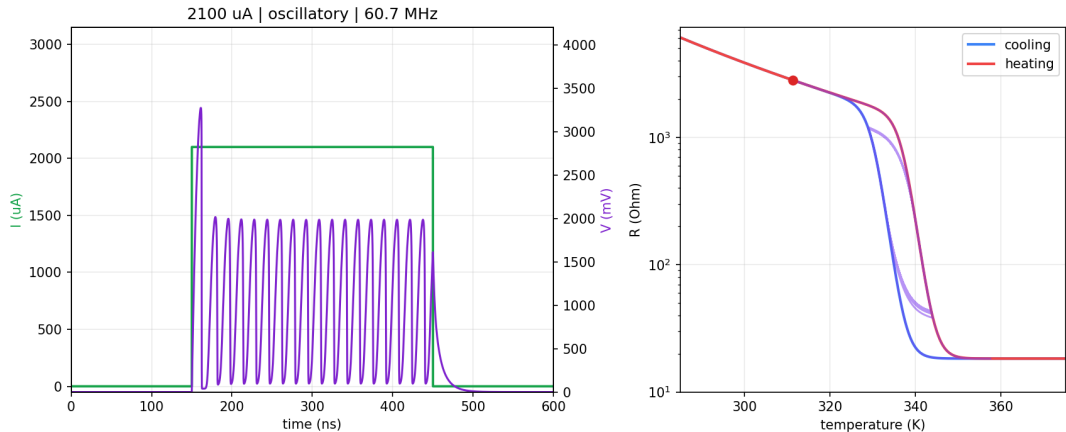
Higher-Frequency Oscillation: 57.1 MHz



1900 μA , $C = 4 \text{ pF}$, $C_{th} = 1 \text{ pJ K}^{-1}$: 57.1 MHz, $V_{pp} \approx 2.08 \text{ V}$, and $E_{\text{cycle}} \approx 39.1 \text{ pJ}$. The trajectory repeatedly closes a stable minor loop.

► [Open high-frequency current-sweep animation](#)

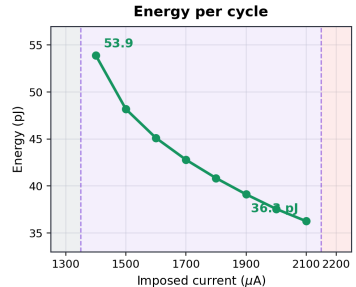
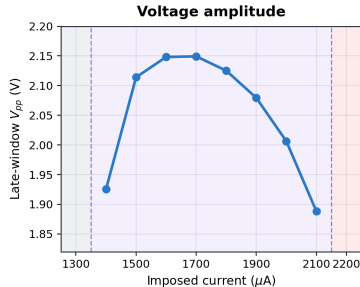
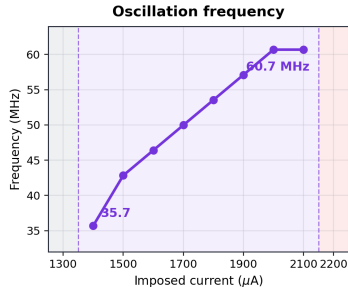
Upper Edge of the Oscillatory Window



At 2100 μA , frequency reaches 60.7 MHz, $V_{pp} \approx 1.89 \text{ V}$, and $E_{\text{cycle}} \approx 36.3 \text{ pJ}$. This is the last sustained point before metallic lock at 2200 μA .

Mapped Operating Window

Quantitative response inside the sustained oscillatory domain



Shading: insulating equilibrium | sustained oscillations | metallic lock

Window: 1400 μA to 2100 μA . Frequency increases while cycle energy decreases; voltage amplitude remains approximately 2 V.

$C = 4 \text{ pF}$, $C_{th} = 1 \text{ pJ K}^{-1}$, $S_e = 0.10 \text{ mW K}^{-1}$; current applied from 150 ns to 450 ns.

What Is Established, and What Is Not

Supported by the simulations

- The original Yuanhang hysteresis mechanism can oscillate under ideal current drive.
- Low, oscillatory, and metallic-lock regimes emerge without adding a phase-delay state.
- The paper-frequency analog reaches 35.7 MHz to 60.7 MHz.
- Serial and vectorized traces agree; representative periods converge with smaller Δt .

Not yet an absolute experimental fit

- The simulated current window is higher than the measured one.
- Cycle energy is ~ 36 pJ to 54 pJ, not the paper's ~ 1 pJ to 2 pJ.
- C , C_{th} , S_e , and source non-ideality need transient calibration.
- The model is phenomenologically corroborative, not parameter-identical.

From Mechanism Demonstration to Specimen Calibration

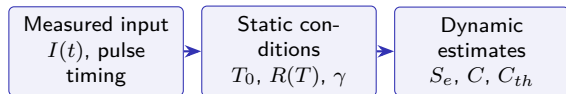
What the July study established

The preserved Yuanhang hysteresis mechanism can generate robust current-driven relaxation oscillations in a verified numerical domain.

What it did not establish

The illustrative parameter set did not reproduce the measured current window, voltage amplitude, or energy scale.

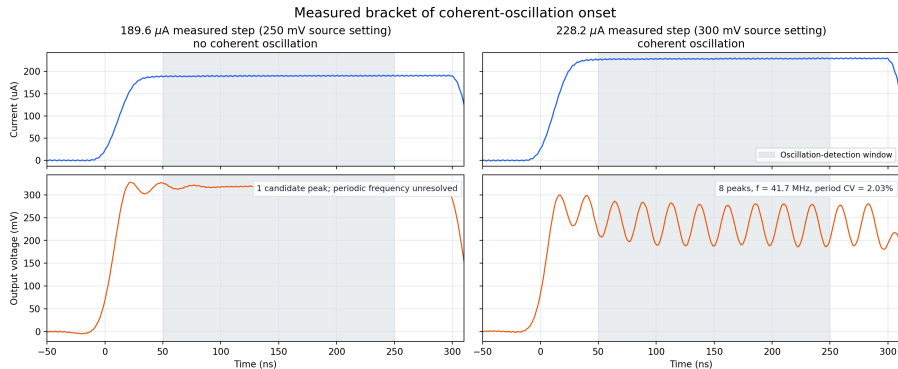
Specimen-specific inference chain



- 1 Estimate each quantity from the most direct available evidence.
- 2 Record whether it is fitted, adopted, bounded, or borrowed.
- 3 Freeze one parameter vector before replaying the full current sweep.

Numerical evidence: reviewed bundles in `public_jobs/`; no per-trace tuning in the final replay.

Input Current $I(t)$: Replay the Measured Waveform



Measured operating input

$$I_{\text{model}}(t) = I_{\text{measured}}(t)$$

The 250 and 300 mV source settings correspond to measured current steps of 189.6 and 228.2 μA . The pulse lasts 300 ns, with a 26–27 ns rise and an 18–23 ns fall.

Professor-supplied numerical traces; archived onset analysis ec6ec4.

Ambient Temperature T_0 : Adopted Experimental Condition

Role in the thermal balance

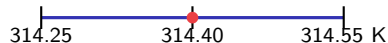
$$C_{th} \frac{dT}{dt} = P - S_e(T - T_0)$$
$$\Delta T = T - T_0$$

The TIA sweep did not contain an independent thermometer record. We therefore adopted the ambient range reported for the related measurements rather than fitting T_0 from the electrical traces.

Adopted value

$$T_0 = 314.4 \text{ K}$$

reported range: 314.25–314.55 K



Interpretation

This is an adopted boundary condition, not a specimen-specific fit. Its applicability to the exact TIA run remains an experimental check.

Ambient range: Gildor et al., arXiv:2604.04594v1; inference status recorded in the research handoff.

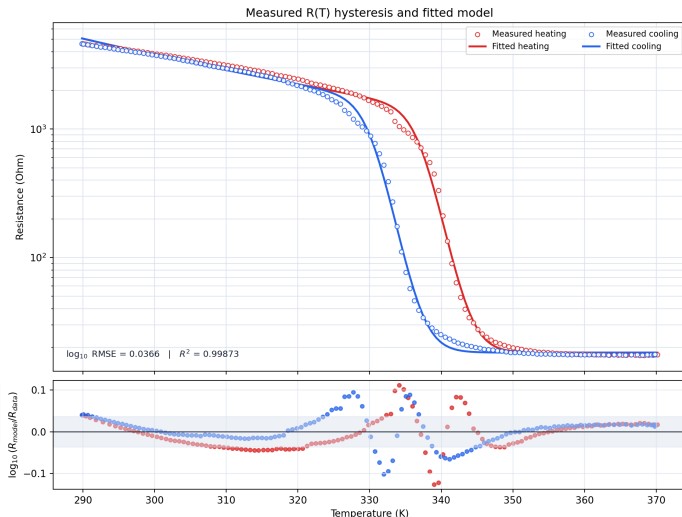
Resistance Parameters: Fit the Measured Major Loop

$$\min_{\theta} \sum_i [\log_{10} R_{\text{model}}(T_i; \theta) - \log_{10} R_i]^2$$

Parameter	Specimen	Yuanhang
R_0 (Ω)	0.7989	0.00536
E_a/k_B (K)	2536.97	5220.47
R_m (Ω)	18.215	1286.32
T_c (K)	333.494	332.806
w (K)	6.882	7.194
β (K^{-1})	0.2992	0.2528

Fit quality

$$R^2 = 0.99873 \quad \text{RMSE}_{\log_{10} R} = 0.03663$$



264 measured points from sample 100425 chip1 gap3; reviewed fit bundle 0849a9.

Minor-Loop Memory γ : Not Identified by the Major Loop

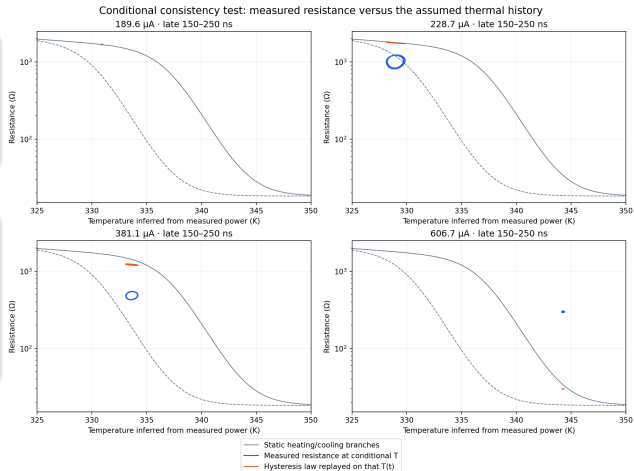
What the measurement determines

One complete heating branch and one complete cooling branch constrain R_0 , E_a , R_m , T_c , w , and β .

What γ controls

γ changes the reversal-dependent minor-loop return. A single major loop contains no independent family of partial reversals, so it cannot determine γ .

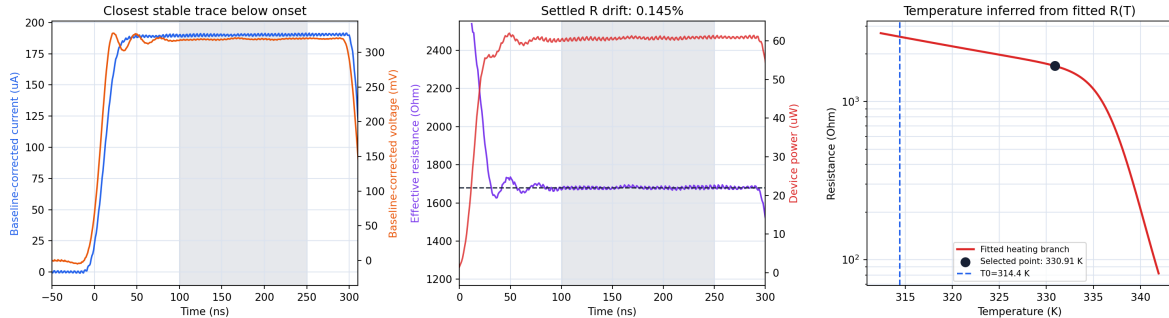
$$\gamma = 0.95627 \text{ (borrowed from Yuanhang)}$$



Bundle fa4b66: scanning $\gamma = 0.157$ to 2.0 did not repair the driven mismatch. A specimen value requires measured minor loops.

Environmental Conductance S_e : Use the Last Stable Trace

Environmental thermal-conductance estimate from numerical waveforms



At the settled pre-onset point, $dT/dt \approx 0$:

$$S_e = \frac{P_*}{T_* - T_0} = \frac{60.658 \mu\text{W}}{330.905 - 314.4 \text{ K}}$$

Selected record: 190.162 μA settled current; resistance drift 0.145%; reviewed bundle 761640.

Result

$$S_e = 3.675 \mu\text{W/K}$$

conditional range 3.434–4.085 $\mu\text{W/K}$

Electrical Capacitance C : A Timing Bound, Not a Detection

Use the complete current balance

$$C \frac{dV}{dt} = I_{\text{in}} - \frac{V}{R(T)}$$

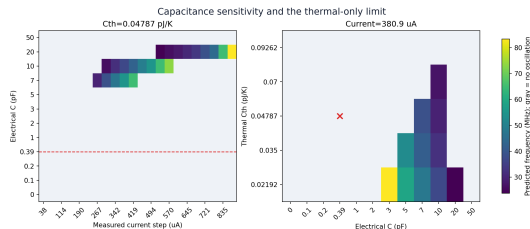
The early, non-switching edges leave no resolved positive capacitive current. The constrained optimum is $C = 0$. With a 1 ns timing resolution and $R(T_0) = 2.570 \text{ k}\Omega$,

$$C \lesssim \frac{1 \text{ ns}}{2.570 \text{ k}\Omega} = 0.39 \text{ pF}.$$

Adopted modeling value

$$C = 0.39 \text{ pF} \quad \text{conservative upper bound}$$

Electrical-edge analysis and frozen-model sensitivity bundle eefab7.



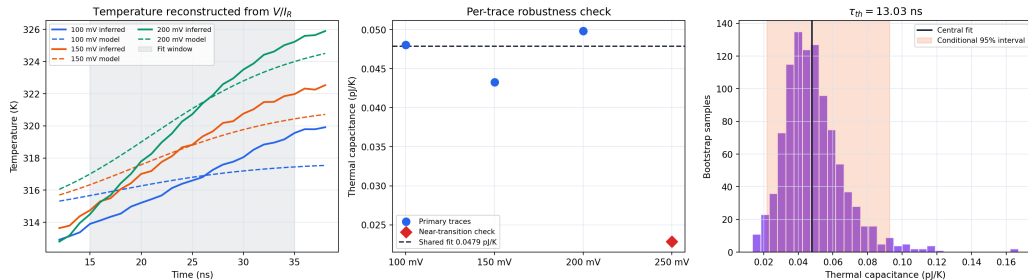
The red dashed line marks 0.39 pF. In the frozen forward study, oscillations appeared only at several-pF values, outside the timing bound.

Why 19.8 pF was rejected

$I/(dV/dt)$ neglects the resistive current. It would imply $RC = 50.9 \text{ ns}$, a voltage lag absent from the measured edge.

Thermal Capacitance C_{th} : Fit a Shared Heating Time

Thermal time constant and capacitance from nonswitching heating edges



$$I_R = I - C \frac{dV}{dt}, \quad T_* = R_{\text{heat}}^{-1} \left(\frac{V}{I_R} \right), \quad C_{th} \frac{dT}{dt} = V I_R - S_e (T - T_0)$$

One C_{th} fits the 76.2, 114.2, and 152.2 μA heating edges from 15–35 ns.
The 189.6 μA near-transition trace is a sensitivity check.

Result

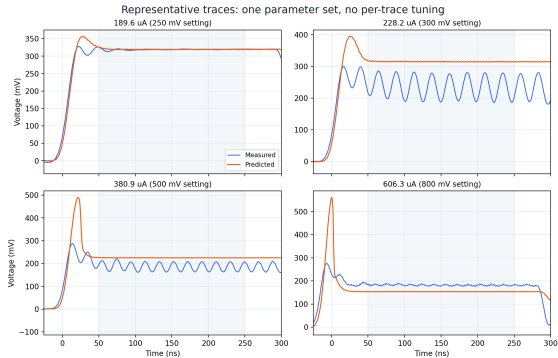
$$C_{th} = 0.047873 \text{ pJ/K}$$

$$\tau_{th} = C_{th}/S_e = 13.026 \text{ ns}$$

conditional range 0.021918–0.092624 pJ/K

Shared transient fit with $C = 0.39$ pF; reviewed bundle aa2469.

Blind Replay: One Frozen Parameter Set Across All Currents



Result

The stable 189.6 μA record is reproduced closely. The frozen model predicts no sustained oscillation in any of the 11 measured oscillatory records and misses the voltage level at intermediate and high currents.

No per-current tuning; frozen specimen validation bundle eefab7.

Can Joint Fitting Rescue the Existing Model?

Question

Can one shared parameter vector explain both the same-device static $R(T)$ curve and the measured current-driven voltage dynamics?

Eleven free quantities

- All six major-loop parameters:
 $R_0, E_a/k_B, R_m, T_c, w, \beta$.
- Minor-loop curvature γ .
- C, S_e, C_{th}, T_0 .

Two controlled searches

- Strong static-data penalty: preserve $R(T)$.
- Weak penalty: diagnostic tradeoff with dynamics.
- Same waveform loss, bounds and starting population.

Nine representative current records train the model; thirteen are excluded from optimization. All were examined in earlier research, so this is not a new blind test.

Recipe: `experiments/current/specimen_joint_inference.toml`. No change to the physical equations or hysteresis.

A Bounded Search Across Different Dynamical Regimes

- 1 Normalize the search: log coordinates for positive scales, $\tau_{th} = C_{th}/S_e$, and insulating resistance at 315 K instead of R_0 .
- 2 Combine historical fits, nearby perturbations and a space-filling sample; reject severe static-data disagreement before running the simulator.
- 3 Run differential evolution with 24 candidates and three generations per objective, then at most 24 Powell evaluations per objective.
- 4 Cache repeated simulations; verify the winners and historical references on all 22 currents at 0.025, 0.0125 and 0.00625 ns.

Budget, not a global optimum

At most 240 objective calls. A short search can find an improvement; failure to fit does not prove that no parameter vector exists.

Search timestep: 0.05 ns. Full recorded pre-pulse history retained; unused post-250 ns tail omitted.

Fit Persistent Features Alongside the Static Curve

$$L = L_{\text{mean}} + 3L_{\text{periodic amplitude}} + L_{\text{robust amplitude}} + 0.5L_f + \lambda_R \left(\frac{\text{RMSE}_{\log_{10} R}}{0.05} \right)^2.$$

- Compare amplitude and mean in four 50 ns windows, from 50 to 250 ns.
- Amplitude uses log ratios with a 3 mV floor; mean-voltage scale is 20 mV.
- Frequency uses a 10 MHz scale and a smooth amplitude/coherence gate; missing cycles are still penalized by their amplitude error.
- $\lambda_R = 4$ preserves the static curve; $\lambda_R = 0.15$ explores the cost of allowing it to move. Both reject static RMSE above 0.35 decades.

Separate acceptance check

Report late mean, amplitude, frequency and four-window persistence independently. Loss weights are engineering choices, not experimental confidence intervals.

Joint Fitting Improves a Score, but Not Every Requirement

Shared candidate	Static RMSE (decades)	Mean RMSE (mV)	Amplitude MAE (mV)	Recovered /11
Frozen estimates	0.0366	45.3	44.3	0
Prior anchored, replayed	0.0426	50.9	140.1	9
Joint: static-preserving	0.0495	50.5	62.0	4
Joint: dynamic diagnostic	0.1433	60.2	90.3	5

Useful improvement

The static-preserving fit reduces the validation feature score from 15.16 (prior anchored) to 10.07: a 33.5% improvement on the thirteen excluded currents.

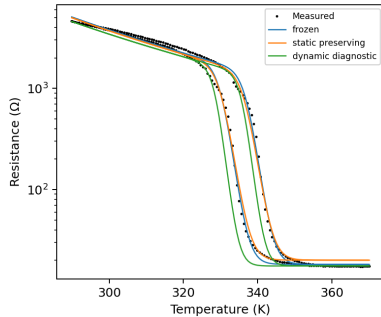
Unresolved tradeoff

Smaller amplitude errors come with fewer sustained oscillators. Mean voltage does not improve over the frozen baseline. No new physical calibration is adopted.

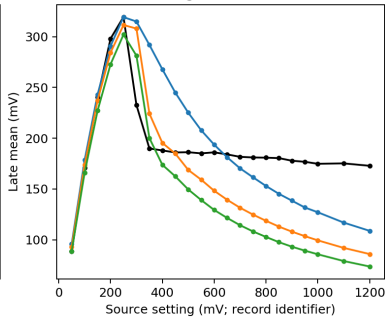
0.00625 ns. Mean error: 22 records. Periodic-amplitude error: the same 11 measured oscillators. Zero false positives. Bundle 6b5d8f.

Static Resistance and Dynamic Waveforms Compete

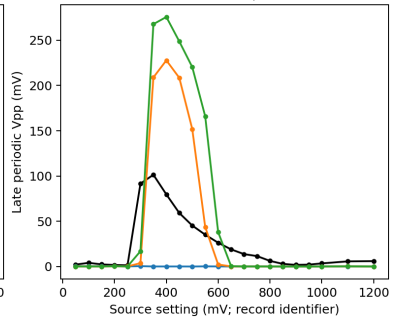
Same-device static $R(T)$



Mean voltage across all records



Oscillation amplitude

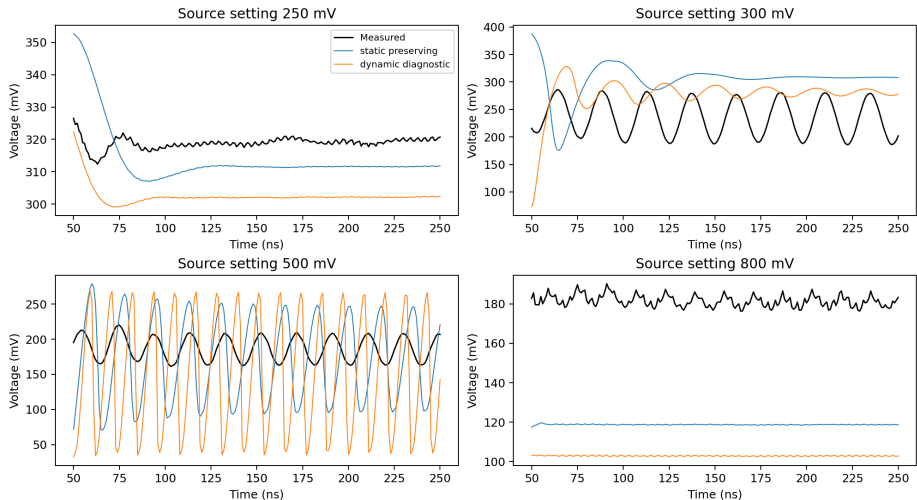


Interpretation

Allowing all resistance parameters to move is not enough, within this budget, to match static $R(T)$, the output plateau and the full oscillation window together.

Plots use one parameter vector per candidate and all 22 records. Horizontal source settings identify records; the simulated input is the measured current waveform.

Representative Waveforms: The Remaining Failures Are Visible



0.00625 ns; no per-current fitting or phase alignment. Settings 250/300/500/800 mV identify measured current steps near 190/228/381/606 microampere.

What the Optimizer Changed

Parameter	Frozen	Static-preserving	Diagnostic
R_0 (Ω)	0.7989	0.6893	0.8620
E_a/k_B (K)	2537.0	2575.1	2485.1
R_m (Ω)	18.22	20.06	17.57
T_c (K)	333.49	333.31	332.18
w (K)	6.882	6.371	7.243
β (K^{-1})	0.2992	0.2800	0.3639
γ	0.9563	0.1757	0.1787
T_0 (K)	314.40	314.53	313.90
C (pF)	0.390	6.806	4.964
S_e ($\mu W/K$)	3.675	3.738	3.519
C_{th} (pJ/K)	0.04787	0.05928	0.03970
τ_{th} (ns), derived	13.03	15.86	11.28

Recurring conflict

Both fits favor electrical capacitance well above the conditional 0.39 pF timing bound and a much smaller minor-loop parameter.

Effective values

These values minimize the chosen objective. They are not independent measurements of device capacitance, heat capacity or filament properties.

All six major-loop quantities were free; numerical values from parameters.csv. Same-device identity was confirmed by Gabriel.

Verification and the Decision After This Search

Last refinement: 0.0125 to 0.00625 ns	Static-preserving	Diagnostic
Maximum late-mean change, all currents	0.34 mV	4.04 mV
Maximum periodic-amplitude change, persistent records	2.52%	1.95%
Maximum frequency change, persistent records	0.16%	0.44%
Persistent records at both fine steps	4	5

- 240 objective calls, 214 unique candidates, 201 search simulations; search wall time 153 s on this machine. Both optimizers exhausted their caps.
- Static-preserving search loss improved 15.1% over its best starting candidate; diagnostic search loss improved 18.7%. These are coarse-search scores.
- Final selection must use fine-step feature errors *and* persistence. No candidate meets the complete experimental target.

Next decision

Preserve these candidates as tradeoff evidence. Calibrate the TIA/device boundary and driven switching law before treating a more flexible fit as physical truth.

Amplitude/frequency changes use records persistent at both fine steps. Full verification: all 22 records at three steps.

Expanded Search: Better Mean Voltage, Remaining Tradeoffs

T_0 was already free over 308–322 K. Added persistence penalties, previous joint-fit seeds and a larger DE/Powell budget; search step halved to 0.025 ns.

Candidate	Static log RMSE	Mean error (mV)	Recovered / 11	Amplitude error (mV)
Prior anchored	0.0426	50.9	9	140.1
Prior joint, stronger static	0.0495	50.5	4	62.0
Expanded, stronger static	0.0489	44.6	7	127.9
Expanded, relaxed static	0.1473	39.8	6	130.6

- 456 objective calls; 400 unique candidates; 399 search simulations; 673 s.
- Stronger-static candidate: $T_0 = 313.09$ K, $C = 4.650$ pF; relaxed candidate: $T_0 = 315.00$ K, $C = 12.334$ pF. Neither hits the ambient bounds.
- Stronger-static recall falls from 8 to 7 at the finest timestep. No candidate resolves the voltage, amplitude and persistence mismatch together.

Bundle 694804. All results at 0.00625 ns; mean-voltage RMSE on 22 records, periodic-amplitude MAE on the same 11 measured oscillators. New loss; historical vectors re-evaluated.

Editable Laboratory: Compare Every Measured Current

Edit and replay

Load original or fitted parameters. Edit the resistance law separately from capacitance, heat capacity, thermal conductance and ambient temperature. Press **Run all measured currents** to use one shared parameter vector.

Inspect

Voltage overlays compare experiment and simulation. A second plot shows simulated resistance versus time. The bottom slider selects the current; optional GIFs animate a time cursor over both plots.

Reproduce

Each run saves numerical traces, metrics, a standalone HTML current slider, per-current GIFs and a parameter recipe. Completed runs are preserved.

Interpretation

Manual replay is exploratory. Check smaller timesteps before interpreting an improved fit. Exported voltage still needs circuit calibration; simulated resistance is not a measured voltage/current ratio.

Start with `neuristor playground`; open `127.0.0.1:8502`.

Frozen Baseline and What the Joint Fit Tests

Quantity	Value	Evidence status
$I(t)$	measured waveform	direct input
T_0	314.4 K	adopted condition
R_m	18.215 Ω	major-loop fit
T_c	333.494 K	major-loop fit
w	6.882 K	major-loop fit
β	0.2992 K ⁻¹	major-loop fit
γ	0.95627	borrowed value
S_e	3.675 $\mu\text{W/K}$	conditional estimate
C	0.39 pF	upper bound
C_{th}	0.047873 pJ/K	conditional fit
τ_{th}	13.026 ns	derived

What the work established

The mechanism implementation works, the static resistance loop is tightly fit, and every dynamic parameter now has an explicit evidence trail.

What the mismatch tells us

The joint search tests whether coupled parameter changes reduce the discrepancy. Physical interpretation still requires the actual VO₂ terminal voltage and branch current, driven minor loops, and an independent thermal response.

Thank you

Backup: Representative Timestep Convergence

Δt (ns)	Period (ns)	Frequency (MHz)
0.015820	123.206	8.116
0.007910	123.135	8.121
0.003955	123.095	8.124

Interpretation

The period approaches a stable value as the timestep is halved. This is the expected signature of a physical solution being resolved more accurately, rather than an oscillation created by the numerical grid.

Representative Yuanhang current-source case: $I = 1640.226 \mu\text{A}$.

Backup: Final Current-Source Standard

$$\begin{aligned} C \frac{dV}{dt} &= I_{in} - \frac{V}{R}, \\ C_{th} \frac{dT}{dt} &= \frac{V^2}{R} - S_e(T - T_0), \\ R(T, \mathcal{H}) &= R_m + R_0 e^{(E_a/k_B)/T} g(T, \mathcal{H}). \end{aligned}$$

- Exactly one VO₂ device/domain and one thermal node.
- Upstream-faithful Yuanhang/Almeida hysteresis and reversal memory.
- Deterministic default $\sigma = 0$; optional stochastic heating remains available.
- Obsolete delayed-phase, double-thermal, and multidomain variants are not active models.

Backup: Reproducibility Trail

- Fitted sample: 100425 chip1 gap3.
- Major-loop fit: 0849a9; conductance estimate: 761640.
- Thermal-capacitance fit: aa2469; frozen validation: eefab7.
- Each reviewed bundle stores resolved inputs, metrics, numerical tables, figures, and a human-readable report.
- Regression tests cover voltage and current models, hysteresis fidelity, convergence, analysis workflows, and archive validation.

AI-use principle

Codex was allowed to automate implementation and search, but successful results were accepted only after model-preservation, physical interpretation, and numerical-convergence checks.